How Browsers Work

From bytes to pixels, in five steps

HTML → DOM
CSS → CSSOM
▸ **Render tree** what is shown ▸ **Layout** geometry ▸ **Paint** pixels ▸ **Composite** stack layers

	<script>	defer	async
Download	stops parsing	in parallel	in parallel
Runs	when reached	after parsing	when ready
Order	kept	kept	any

defer finishes before **DOMContentLoaded**. Both ignore inline scripts.

1 Layout again

```
el.style.width = "300px";
// neighbours move
// layout, paint, composite
```

2 Compositor only

```
el.style.transform = "scale(2)";
// neighbours stay
// can skip layout and paint
```

TRAP What happens here?

```
// in <head>; both log document.body
<script src="a.js"></script>
<script src="b.js" defer></script>
```

```
a.js: null
b.js: <body>
```

The body does not exist yet when a.js runs.

KEY TAKEAWAY

- `CSS` Blocks rendering
- `script` Blocks parsing
- `defer` Waits for the DOM

? Follow-up questions to expect

- `display: none` or `hidden`? → none: no box at all. hidden: space is kept.
- **What is a reflow?** → Layout runs again after a geometry change.
- **DOMContentLoaded or load?** → DOM ready and defer done. load: all resources too.
- **Why is CSS render-blocking?** → Rules can override each other, so it waits.